Photonics in AI Infrastructure: updated case study anchored from 2025Q1

Executive view

The report has been rebased so every portfolio analytic begins in **2025Q1**. That window captures a difficult first quarter, the **5 Oct 2025** ARKA identification point, and the subsequent transition from broad theme discovery into a tighter 2025Q4 basket.

What changed in this revision

1. Analytics now start **1 Jan 2025**.
2. The formal ARKA identification date is **5 Oct 2025**, with first trading-day implementation on **6 Oct 2025**.
3. Top-holdings snapshot now uses the **latest (2026Q1) rebalance weights** — the basket has tightened from 20 names (2025Q1) to 14 (2026Q1).
4. Weight migration is drawn as **quarterly line plots across every rebalance**.
5. Trigger stack is expanded and classified by **type, significance, and public validation**.

FULL VALIDATION WINDOW (REGENERATED)
1.95 Sharpe
+197.9% total return · 47.5% ann vol
1 Jan 2025 → 17 Apr 2026

POST-TRIGGER WINDOW (REGENERATED)
2.96 Sharpe
+101.0% total return · 46.9% ann vol
6 Oct 2025 → 17 Apr 2026

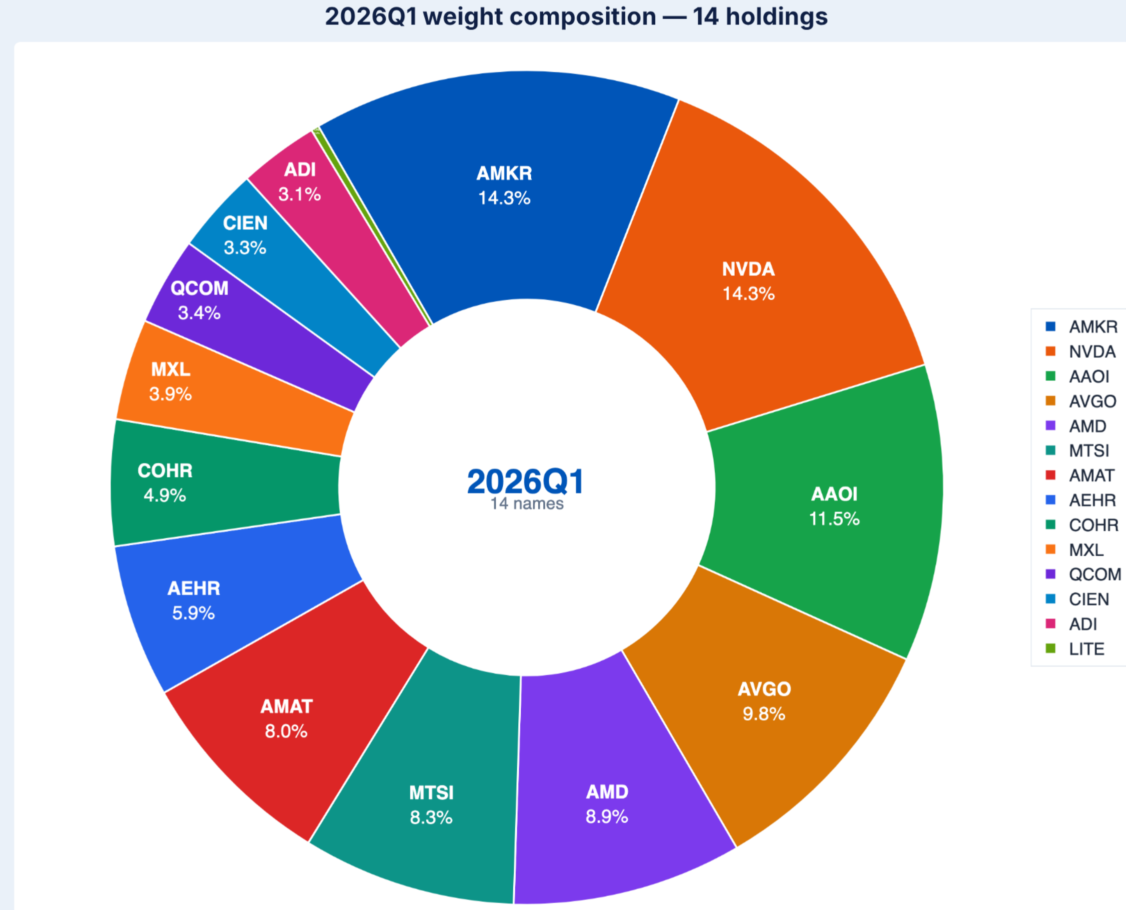
BENCHMARK READ-THROUGH
+62.0% alpha
+2.54 information ratio · R^2 0.72
75.0% of months beat SPY

Indexed return since 1 Jan 2025 (2025Q1 rebased to 100)

2026Q1 holdings — full basket (14 names)

#	Ticker	Weight	Role in the basket
1	AMKR	14.29%	Advanced packaging / assembly
2	NVDA	14.29%	System / switch-platform read-through
3	AAOI	11.52%	Optical transceivers / modules
4	AVGO	9.84%	Switch silicon / CPO platform
5	AMD	8.90%	GPU / systems adjacency
6	MTSI	8.28%	Laser / RF / analog building blocks
7	AMAT	8.03%	Semi equipment — packaging / deposition
8	AEHR	5.93%	Burn-in / test capacity for SiPh
9	COHR	4.88%	Lasers / optical components
10	MXL	3.92%	DSP / analog interconnect
11	QCOM	3.41%	System / connectivity read-through
12	CIEN	3.32%	Optical systems — data-center networks
13	ADI	3.09%	Analog / TIA / driver IP
14	LITE	0.30%	Lasers / pluggable transceivers (tail position)

2026Q1 weight composition — 14 holdings



How the theme was identified — and why 5 Oct 2025 was the right trigger date

The signal was **not** a single headline. ARKA saw a convergence of architecture shifts, validated product roadmaps, reliability milestones, packaging / onshoring commitments, and price confirmation across the optical stack.

2025Q1 replay path with classified triggers and ARKA identification date

What ARKA was seeing

- **Platform architecture:** NVIDIA Spectrum-X / Quantum-X and Broadcom's CPO roadmap turned optics into an AI-fabric problem, not a loose component theme.
- **Module and DSP readiness:** Marvell, MaxLinear / MTSI, and AAOI provided evidence that 800G and 1.6T lanes were moving from roadmap to qualification and order flow.
- **Packaging and domestic capacity:** Amkor and Corning extended the investable stack into assembly, fiber, and U.S. buildout.
- **Reliability and standards:** Broadcom / Meta qualification plus OIF and Microsoft Research reduced the gap between promising technology and deployable infrastructure.

Why the 5 Oct 2025 trigger date matters

ARKA is not looking for one headline. The trigger is the point at which **evidence density, technical validation, and price confirmation** become strong enough to justify a governed factor build.

Classified trigger stack used to frame the signal

Date	Trigger	Type	Significance	Validation	Why it mattered	Ref.
6 Jan 2025	Marvell CPO architecture for custom AI accelerators	Architecture	High	Official product architecture	Made CPO a practical scale-up path beyond intra-rack copper.	P2
18 Mar 2025	NVIDIA Spectrum-X / Quantum-X Photonics launch	Platform launch	Very high	Official launch + ecosystem partners	Moved photonics from component story to system-level AI-factory infrastructure.	P1, P3
31 Mar 2025	Marvell 1.6T silicon photonics light-engine demo	Technical validation	High	OFC demonstration	Confirmed a concrete 1.6T module path and lower-power rack-scale optics.	P4
13 May 2025	Corning qualified for Broadcom Bailly optical infrastructure	Supply-chain validation	High	Official supplier qualification	Showed the fiber/connectivity layer entering the CPO stack, not just chips and DSPs.	P5
8–9 Sep 2025	MOSAIC + Microsoft Research network-wall framing	Research / bottleneck	High	Primary research + engineering blog	Publicly framed the copper-vs-optics trade-off as a binding AI-infrastructure problem.	P6, P7
1 Oct 2025	Broadcom / Meta achieve 1M flap-free CPO link hours	Reliability validation	Very high	Qualification proof	Moved CPO toward production-ready reliability.	P8
5 Oct 2025	ARKA internal identification date	Internal synthesis	Very high	Controller trigger	Elevated the theme into a governed factor build and queued the production run.	I1, I2
6–8 Oct 2025	Amkor Arizona groundbreaking + Broadcom Tomahawk 6 shipping	Capacity + commercialization	Very high	Official capacity + shipment event	Added packaging/onshoring evidence and a shipping CPO switch, improving practical investability.	P9, P10
10 Dec 2025	Applied Optoelectronics first volume 800G order	Commercial validation	High	Company announcement	Converted architecture and qualification into visible order flow.	P11
27 Jan 2026	Corning / Meta up to $6B multi-year optical agreement	Scale validation	Very high	Official strategic agreement	Confirmed optical connectivity spend was becoming a large, explicit data-center budget line.	P12
26 Feb / 9 Mar 2026	OIF AI scale-up publications and COI white paper	Standards / ecosystem	High	Standards-body publication	Reduced standards risk and clarified how AI scale-up optics could industrialize.	P16
2 Mar 2026	NVIDIA / Coherent strategic optics partnership	Capacity + supply security	High	Strategic partnership	Strengthened the domestic manufacturing and long-term supply leg of the thesis.	P13
2 Mar 2026	NVIDIA / Lumentum strategic optics partnership	Capacity reservation	High	Strategic partnership	Added forward-capacity evidence for lasers and optical engines.	P14
12 Mar 2026	Marvell 1.6T DSP and connectivity portfolio expansion	Commercial roadmap	High	Platform commentary	Reinforced that 1.6T monetization was broadening from architecture into portfolio depth.	P15

Trigger significance color-coded; internal dates reference internal materials with official company releases, standards-body announcements, and primary research linked in the References section.

Controller workflow: from prompt to portfolio-ready output

The attached mathematical framework has **two layers** working together. The first is freeform and thesis-aware: ARKA infers the relevant ontology and evidence stack. The second is **compiled and governed**: the system converts the thesis into explicit dimensions, equations, liquidity controls, and an audit pack.

1 Liftoff

PM prompt in natural language. Controller infers supply-chain roles, inclusion tests, and avoid flags. The process is freeform at inception.

↓

2 Mathematical framework

Compile D1–D6 dimensions plus penalty. Apply normalization, weighting, rebalance, imputation, thresholds, and caps. Formal at scoring.

↓

3 Pseudocode

Translate formulas into explicit pipeline: universe → extract → classify → score → select → weight → rebalance → output. Reproducible once compiled.

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4 Orchestration

Specialized agents scan filings, transcripts, patents, standards, news, and market data. Conflicts are reconciled and evidence is linked back to source. Scales across large agent / LLM call graphs.

↓

5 Final file / UI

Emit weights file, rebalance pack, audit trail, rationale manifest, and HTML / PDF / UI factor card. Auditable at output.

Universe selection can start from more than one anchor

PMs can specify the starting universe as **S&P 500**, **Russell 2000**, **theme-only candidates**, or **any union of the three**. The controller is not forced into a single benchmark ontology; it can start broad, narrow, or hybrid depending on the thesis.

Compiled exposure dimensions

Dimension	What it is capturing
D1 Revenue exposure	Explicit SiPh / CPO revenue; product mentions
D2 Speed-node & wins	800G / 1.6T / 3.2T, sample / qualification, hyperscaler wins
D3 Critical components	Laser / modulator IP, vertical stages, component capacity
D4 Packaging capacity	SiPh foundry, 2.5D / 3D / CPO assembly evidence
D5 CPO momentum	Roadmap presence plus win / qualification count
D6 Domestic advantage	U.S. footprint, CHIPS / onshoring / long-term supply
Penalty	Pure-pluggable focus or concentrated China-material risk

Core scoring equation

$$\begin{aligned}\text{Score}_i = [\; & 0.35 \cdot D_1(\text{Revenue exposure}) \\ & + 0.25 \cdot D_2(\text{Speed-node \& wins}) \\ & + 0.20 \cdot D_3(\text{Critical components}) \\ & + 0.10 \cdot D_4(\text{Packaging capacity}) \\ & + 0.05 \cdot D_5(\text{CPO momentum}) \\ & + 0.05 \cdot D_6(\text{Domestic advantage}) \;] \times \text{Penalty}_i\end{aligned}$$

Weights cap-floored, industry- and liquidity-constrained; quarterly rebalance with monthly tradability checks.

Controller pseudocode

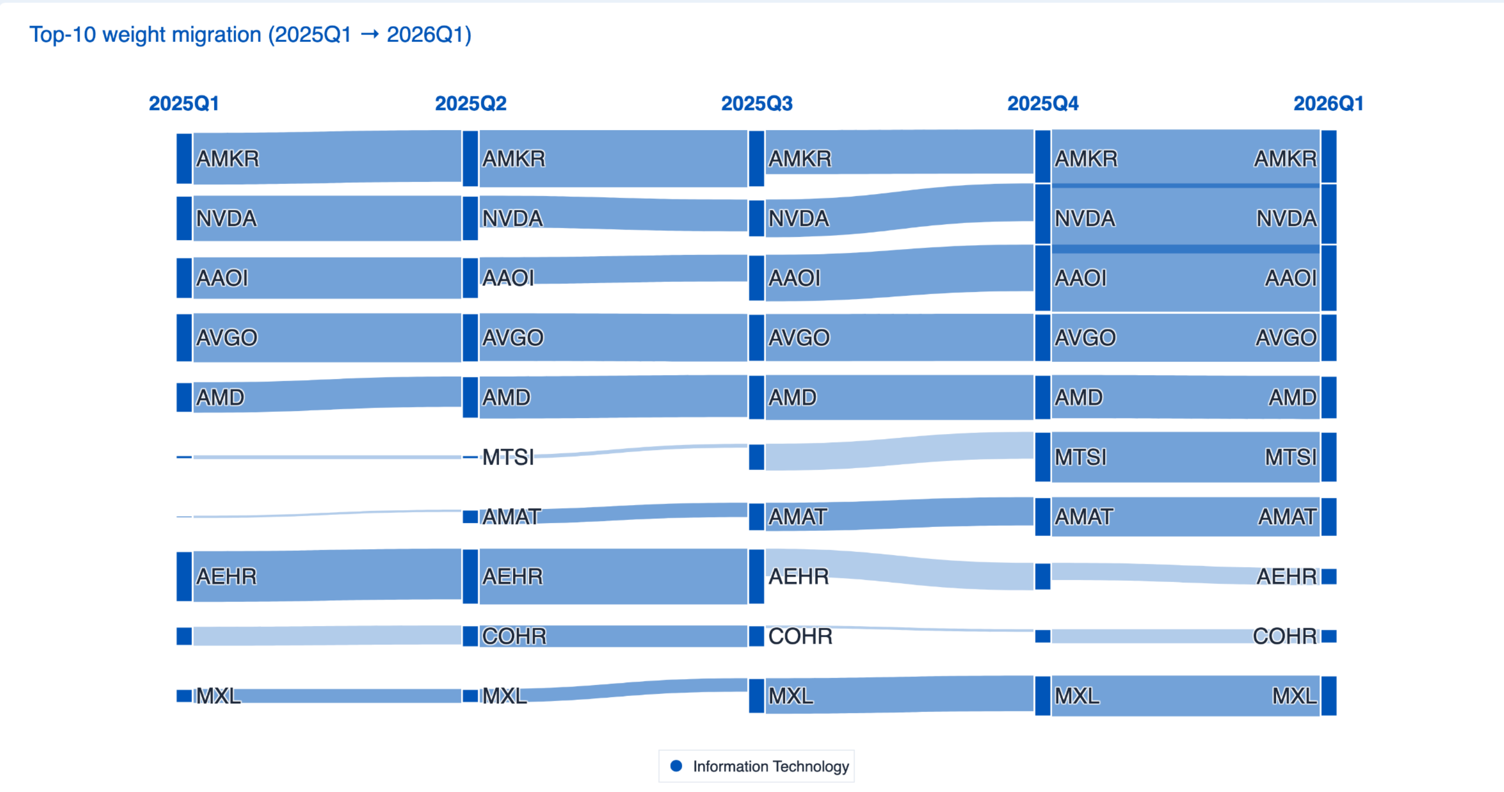
```
Universe   <-  chosen benchmark set(s) or theme-only pool
Filter     <-  cap, ADV, price, listing, tradability thresholds
Evidence   <-  filings + transcripts + patents + standards + price validation
Classify   <-  revenue, speed nodes, wins, capacity, onshoring, avoid flags
Score      <-  normalize -> winsorize -> aggregate dimensions -> apply penalty
Select     <-  rank, buffer, exclude avoid-list names
Weight     <-  S² subject to single-name, industry, and liquidity caps
Maintain   <-  quarterly rebalance + monthly tradability maintenance
Output     <-  weights file + audit trail + report + UI card
```

Freeform at inception → formal at scoring → deterministic at portfolio → auditable at output

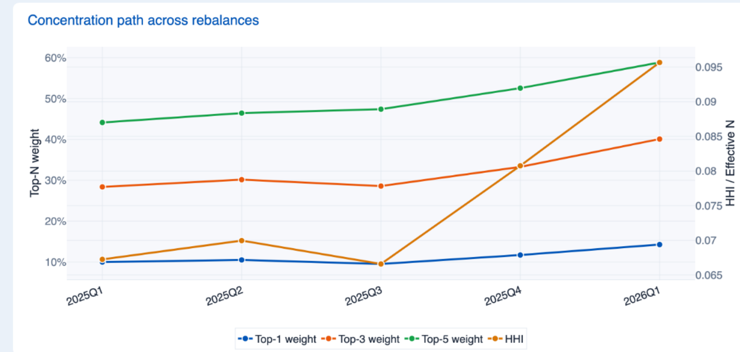
Basket construction: explainable weight changes, not theme drift

The rebased window shows the basket tightening from a broader 2025Q1 discovery set into a higher-conviction **2026Q1** expression (**20 names → 14 names**). The key point is **not concentration by itself** — it is that weight change follows **public validation, production-readiness, and liquidity discipline**. The basket is moving toward the layers where photonics is actually monetized or bottlenecked: **system platforms, optical modules, packaging scale, and qualification capacity**.

Quarterly weight migration of 2026Q1 constituents



Concentration path since 2026Q1



What the weight changes are saying

- **Chart scope:** drawn only for names that survive into the **2026Q1 basket** — read it as a **tightening of conviction**, not a full holdings census.
- **Weight now sits in AMKR, NVDA, AAOI, AVGO** (packaging / system-platform / modules / switch-silicon) — where public validation has moved from architecture and roadmap into **deployment, order flow, and scaled manufacturing**.
- **AMD, MTSI, AMAT** — important enabling positions; **AEHR, COHR, MXL, QCOM, CIEN, ADI, LITE** — smaller read-through exposures across test, analog, laser, DSP, and connectivity layers.
- **Not theme drift**: the basket is reallocating toward the **hardest-to-replicate bottlenecks** under explicit portfolio and liquidity constraints.

Representative supply-chain map and why names matter

Stack layer	Representative names	Why weight moved / why it stayed
System / switch platform	**NVDA, AVGO, AMD, QCOM**	Where optical adoption is monetized at the fabric level: switch silicon, GPU / system adjacency, and connectivity platforms with direct pull-through into optical density and AI-cluster scale.
Modules / connectivity	**AAOI, CIEN, LITE**	Direct monetization of optical engines, transceivers, and interconnect links for data-center scale-up and scale-out.
Lasers / DSP / analog	**MTSI, MXL, COHR, ADI**	The driver, TIA, DSP, laser, and analog layer that makes higher lane rates, coherent optics, and CPO architectures commercially usable.
Packaging / assembly	**AMKR, AMAT**	Advanced packaging, assembly, and manufacturing leverage once the architecture moves from design win to scaled deployment.
Test / qualification	**AEHR**	Burn-in and qualification capacity on the path from optical evaluation to volume production.

Interpretation

The factor does **not** buy the entire AI stack. It reallocates toward the bottlenecks that are **hardest to replicate quickly** and **most directly validated by deployment**: **system platforms, optical modules, packaging readiness, and qualification capacity**. The result is a tighter basket whose largest weights sit closer to **commercialization and volume ramp**, while smaller positions preserve read-through to **analog, DSP, laser, and connectivity** layers. In that sense, the updated basket reads as **evidence convergence under portfolio rules**, not thematic drift.

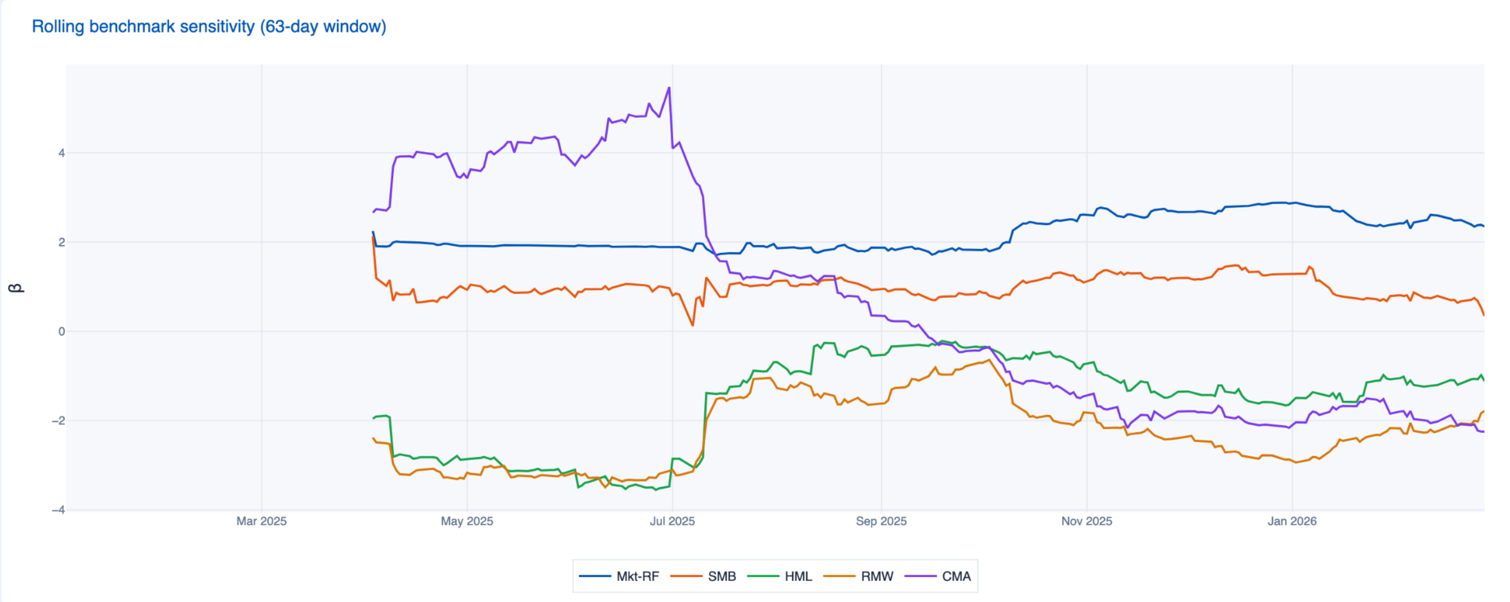
Performance diagnostics after rebasing to 2025Q1

The updated window captures one weak quarter followed by four periods of strong active recovery. That makes the path more credible for PM review than a single headline endpoint.

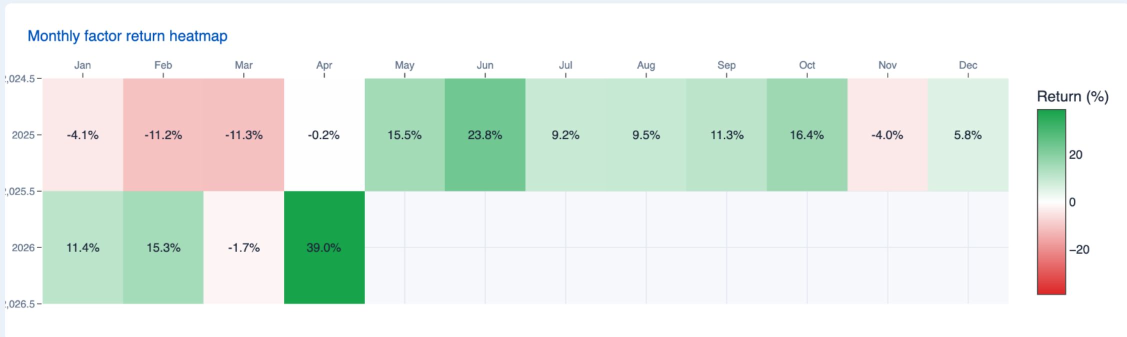
Quarterly active-return profile

Period	Factor	SPY	Active
2025Q1	−24.4%	−4.6%	−19.9%
2025Q2	+42.7%	+10.5%	+32.3%
2025Q3	+33.1%	+7.8%	+25.3%
2025Q4	+18.3%	+2.4%	+15.9%
2026 YTD*	+63.7%	+0.6%	+63.1%

Rolling benchmark sensitivity (63 trading days)



Monthly return heatmap (factor only)



POSITIVE MONTHS
62.5%
2025Q1 to 13 Apr 2026

OUTPERFORMANCE MONTHS
75.0%
Share of months beating SPY

MONTHLY UPSIDE CAPTURE
5.51x
Capture in up-SPY months

MONTHLY DOWNSIDE CAPTURE
0.44x
Capture in down-SPY months

Updated read-through

The updated window still shows a **pro-risk factor**, but with two important institutional features. First, most of the active return arrives **after** the October 2025 identification point. Second, the path includes a negative first quarter, which makes the later recovery more informative than a monotonic backtest.

BEST FULL MONTH
Jun 2025 +23.8%
Best complete month in-window

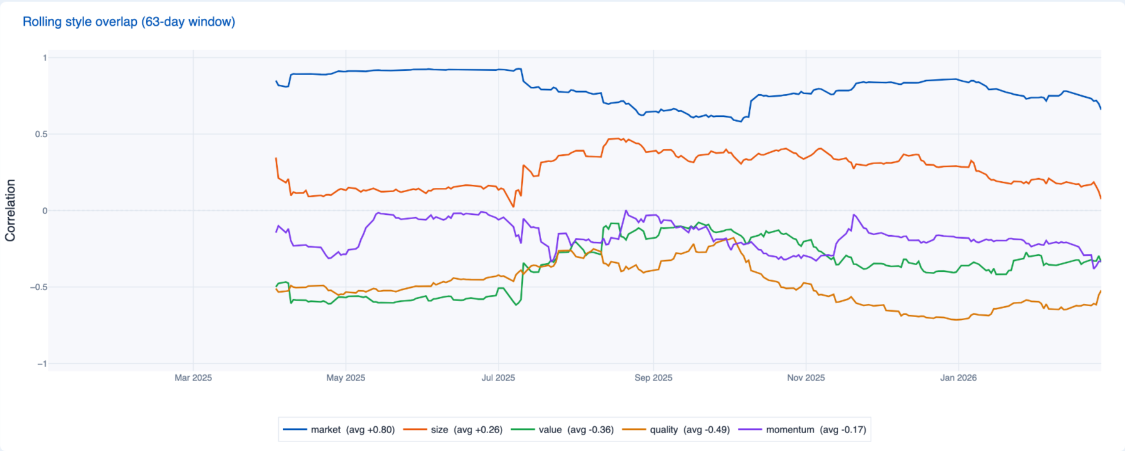
WORST FULL MONTH
Mar 2025 −11.3%
Worst complete month in-window

*2026 YTD is partial through 13 Apr 2026. Monthly heatmap shows factor returns only; quarterly table reports factor, SPY, and active return.

Distinctiveness, style read-through, and where returns came from

The updated window still reads as pro-risk, but it is **not just a benchmark clone**. The point is to distinguish market validation from stack-specific alpha.

Average rolling style overlap since 2025Q1

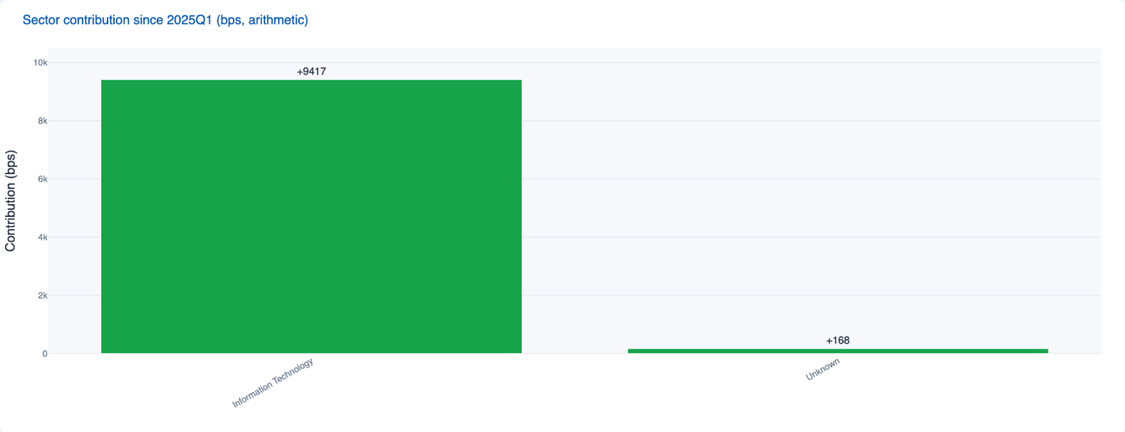


FF5A archive reference (full attached history)

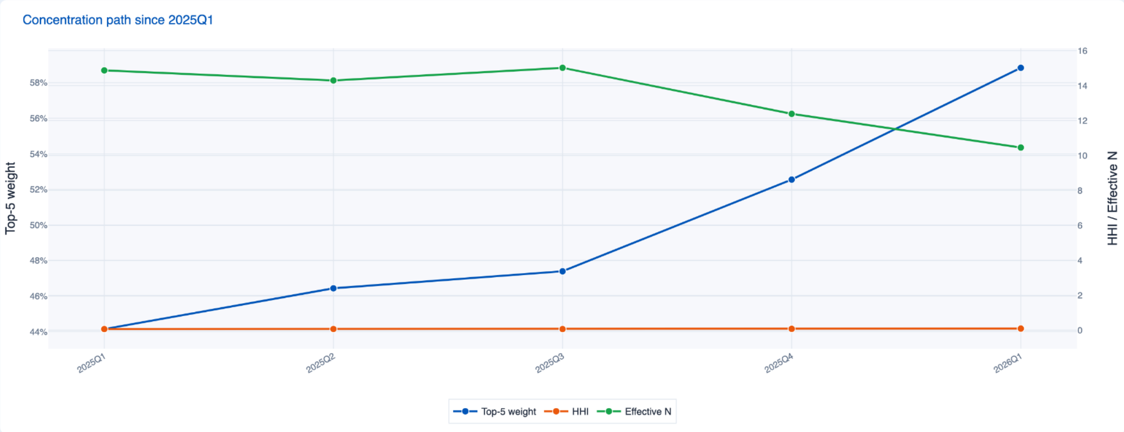
Archive FF5A regression remains useful as a read-through on the longer history. Market beta is **1.50**, SMB **0.50**, HML **−0.68**, RMW **−0.22**, CMA **0.60**. Archive annualized alpha is **51.6%** with **t-stat 2.84**; model fit is **R^2 0.373**.

Factor	Beta	t-stat	Archive contribution
Mkt-RF	+1.50	26.83	+9.8
SMB	+0.50	5.36	−5.0
HML	−0.68	-6.19	+1.9
RMW	−0.22	−1.76	−1.6
CMA	+0.60	3.71	+1.3

Contribution since 2025Q1 (arithmetic bps)



Concentration path since 2026Q1



Updated interpretation

Average rolling correlation since 2025Q1 is still highest to **market (+0.69)**, but overlap is meaningfully lower to **value (−0.34)**, **quality (−0.46)**, and **investment (−0.01)**. By the end of the window, the latest rolling market correlation is down to **+0.33**, showing the basket is not statically tied to broad beta.

Attribution and concentration

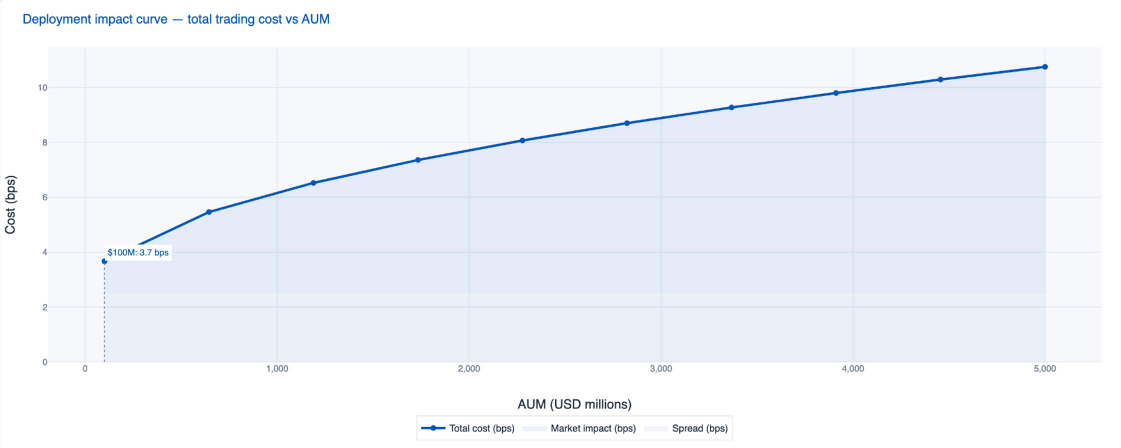
Since 2025Q1, the biggest positive arithmetic contributors are **AAOI, AEHR, LITE, AMKR, and MTSI**. Principal laggards are **UCTT, SMTC, and ENTG**. Top-5 weight rises from **44.1% in 2025Q1** to **52.6% in 2025Q4**, while holdings count falls from **20 to 16**. That tightening is consistent with evidence convergence, not random drift.

Archive FF5A numbers on this page come directly from the attached analytics archive. Updated 2025Q1-window diagnostics are drawn from the rebased return and holdings panels.

Deployment scale and what this says about ARKA

The last test is implementation review. The attached archive still tags the basket as **LARGE**, with roughly **5.9%** average participation and **8.2 bps** total trading cost at a **$100M** deployment.

Illustrative deployment impact from the archive



Deployability snapshot

Metric	Archive view
Capacity tier	**LARGE**
Estimated capacity	2422.1 archive units
Average daily volume	4494.9
Limiting factor	Market impact
Market impact	5.36 bps
Spread cost	2.50 bps
Total cost	**8.21 bps**

Most capacity-sensitive current archive names

Ticker	Weight	ADV (USD mm)	% ADV at $100M
AEHR	5.93%	15.9	37.2%
MXL	3.92%	19.7	19.9%
AMKR	14.29%	132.2	10.8%
AAOI	11.52%	156.7	7.4%
MTSI	8.28%	186.5	4.4%

1 Not a static theme library

The prompt is freeform. ARKA first generates the relevant ontology, evidence stack, inclusion logic, and avoid flags for the thesis at hand.

2 Formal where it matters

Once the theme is compiled, the factor is constrained by explicit math: normalization, weighting, caps, buffers, turnover guards, and liquidity rules.

3 Self-adapting rather than fixed

As evidence migrates from megacap incumbents to niche bottlenecks, the basket can reallocate accordingly. That differs from a static ETF-like label.

4 Output is portfolio-ready

The endpoint is not an interesting narrative. It is a weights file, rebalance pack, rationale manifest, and a UI / report artifact PMs can review.

Freeform at inception. Disciplined in implementation. Repeatable across emergent supply-chain, geopolitical, and technology shocks.

Capacity, participation, and trading-cost estimates on this page come directly from the archive's liquidity section and are best read as implementation diagnostics rather than promises.